

Do **not** write solutions on this page.

10. [Maximum mark: 18]

Consider the expression $\frac{1}{1+x^2}$, where $|x| < 1$.

(a) (i) Show that the first four terms in the binomial expansion of $\frac{1}{1+x^2}$ are $1 - x^2 + x^4 - x^6$. [3]

(ii) Hence, find an approximation for $\int \frac{1}{1+x^2} dx$ up to and including the term in x^7 . [2]

Now consider the expression $\frac{1}{\sqrt{1-x^2}}$, where $|x| < 1$.

(b) (i) Expand $\frac{1}{\sqrt{1-x^2}}$ up to and including the term in x^4 .

(ii) Hence, find a polynomial expression for $\int \frac{1}{\sqrt{1-x^2}} dx$ up to and including the term in x^5 . [5]

The expression $\frac{25}{48} + \frac{k}{1280}$, where $k \in \mathbb{Z}^+$, can be used to approximate $\arcsin \frac{1}{2}$.

(c) Use the result from part (b)(ii) to find the value of k . [4]

(d) Hence, find an approximation for π in the form $\frac{p}{640}$ where $p \in \mathbb{Z}^+$. [4]

